

Antiferromagnetic Superexchange and Oxygen Vacancies in Co-Doped V_2O_5 Nanoparticles: A DFT+U and Experimental Study

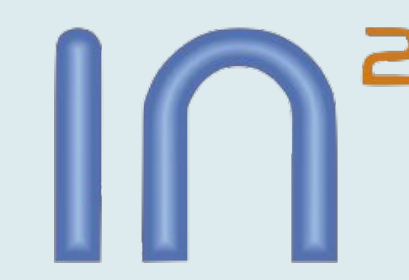
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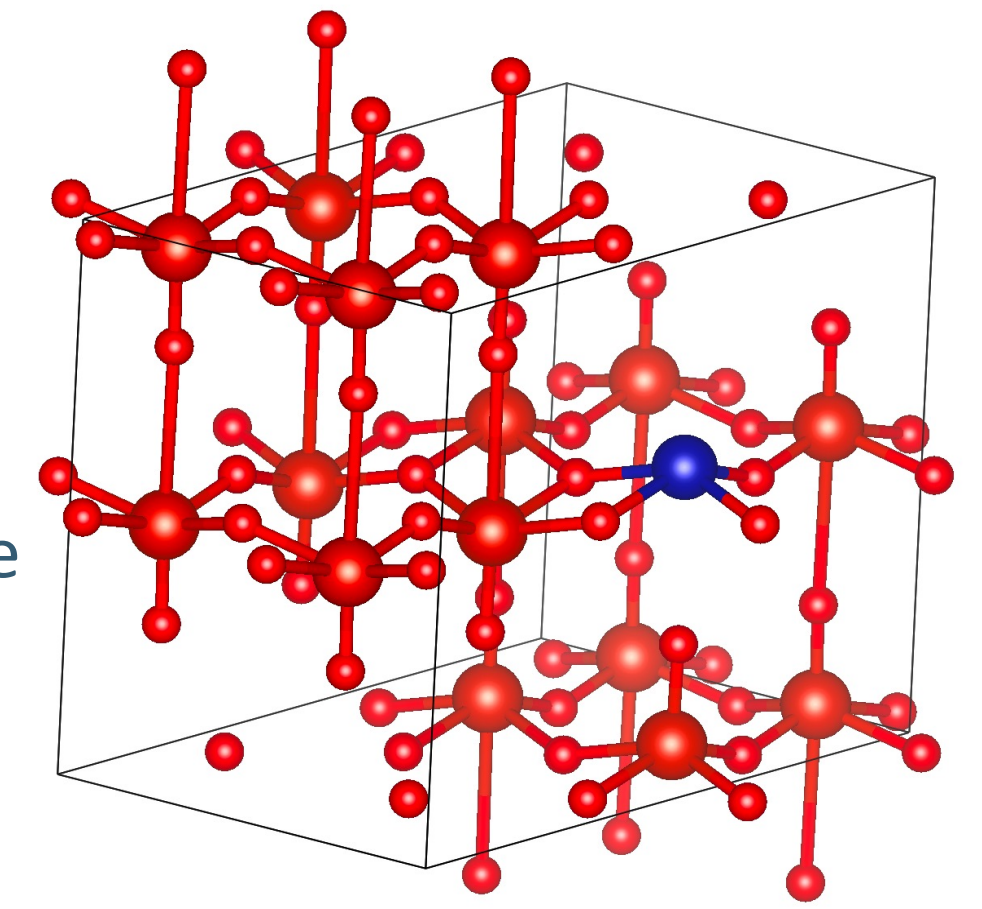
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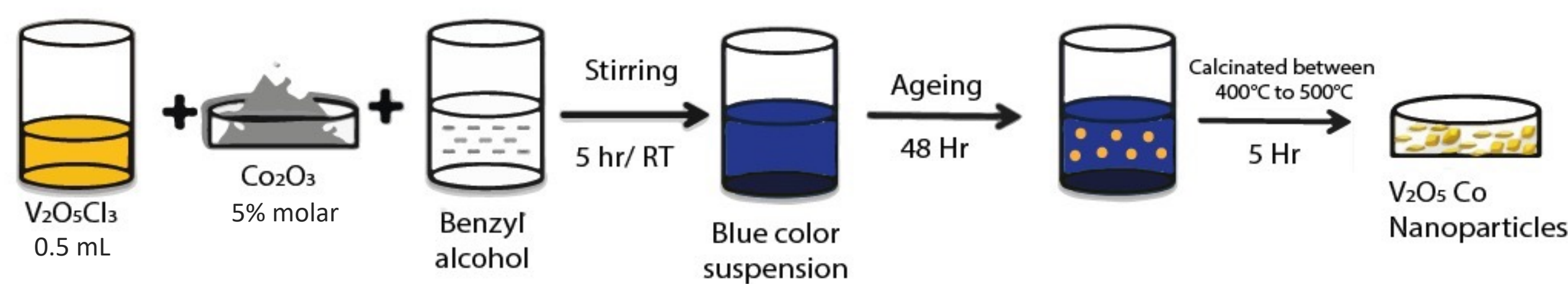
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Introduction

- V_2O_5 nanoparticles are promising for photocatalysis and neuromorphic computing due to tunable electronic properties under thermal processing
- Previous work [1] on undoped V_2O_5 established:
 - Optical band gap reduction from 2.20 to 1.18 eV with increasing calcination temperature (400-500 °C)
 - Ferromagnetic-like behaviour arising from spin-polarized interband states at O1 (vanadyl) vacancies
 - DFT+U confirmation that structural relaxation alone cannot account for the gap reduction (O1 vacancies required)
- Open question: What happens when a magnetic dopant (like Co^{2+}) is introduced into this vacancy-rich lattice? Does it enhance or suppress the intrinsic defect magnetism?
- This work: We synthesize 5% Co-doped V_2O_5 nanoparticles and observed a suppression of the net magnetization relative to undoped samples
- Proposed mechanism: Antiferromagnetic superexchange coupling between Co dopants and vacancy induced V^{4+} polarons

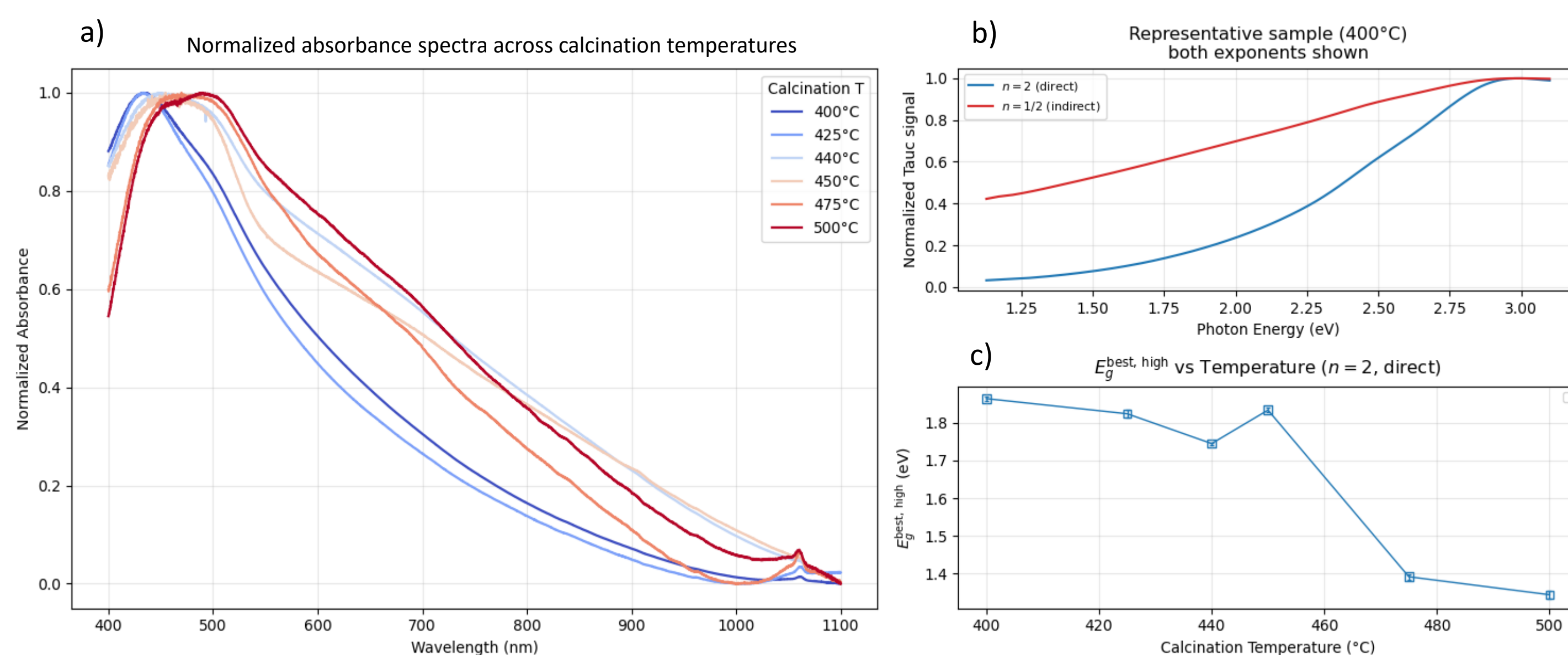


Synthesis

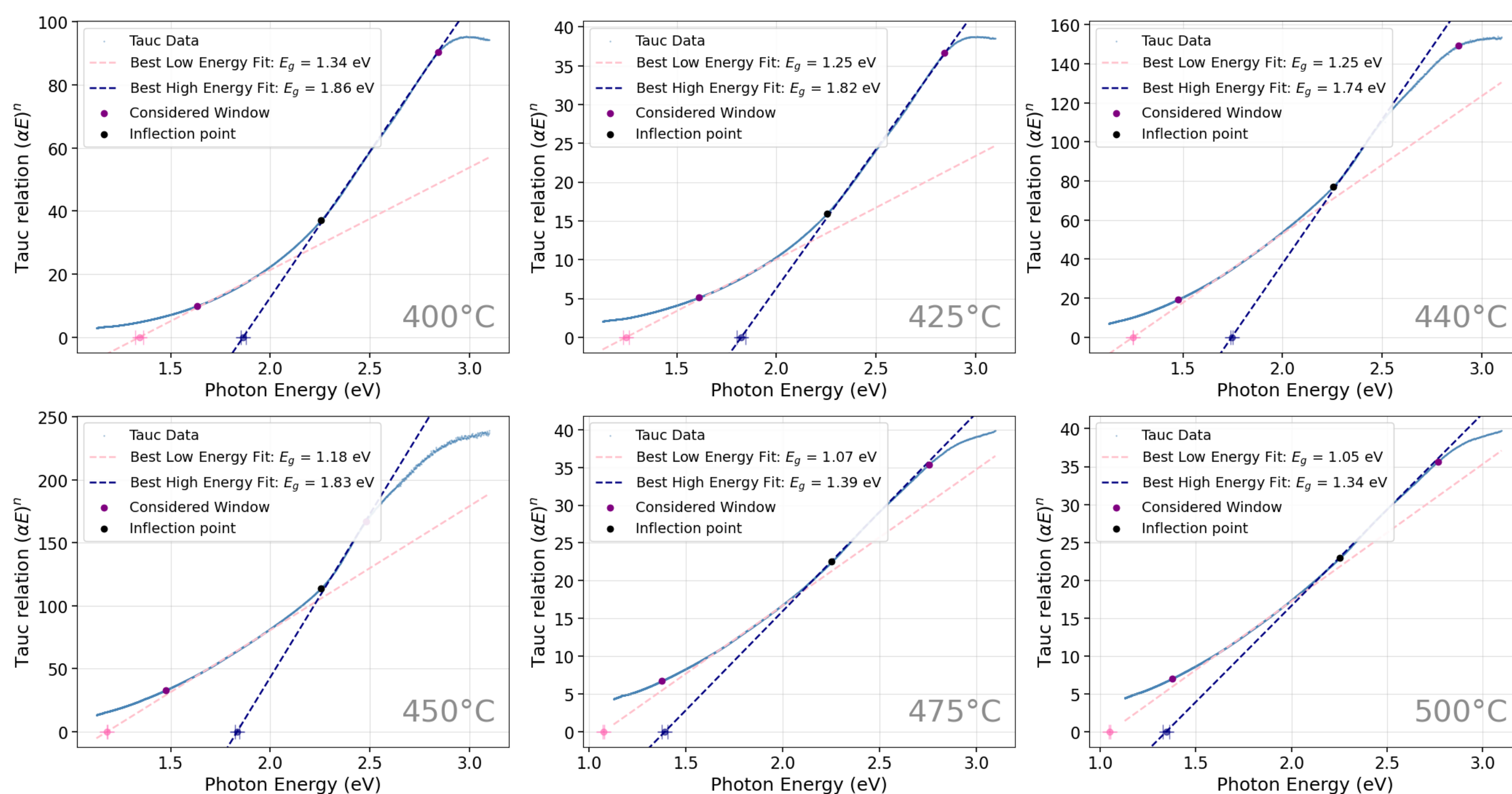


- Non-aqueous sol-gel route [1]: 0.5 mL $VOCl_3$ + 20 mL benzyl alcohol.
- Calcine 5 h at $T_{cal} = 400, 425, 440, 450, 475, 500$ °C.
- For Co-doped samples: 5% molar $CoCl_2$ added to precursor before gelation.

Optical Absorption

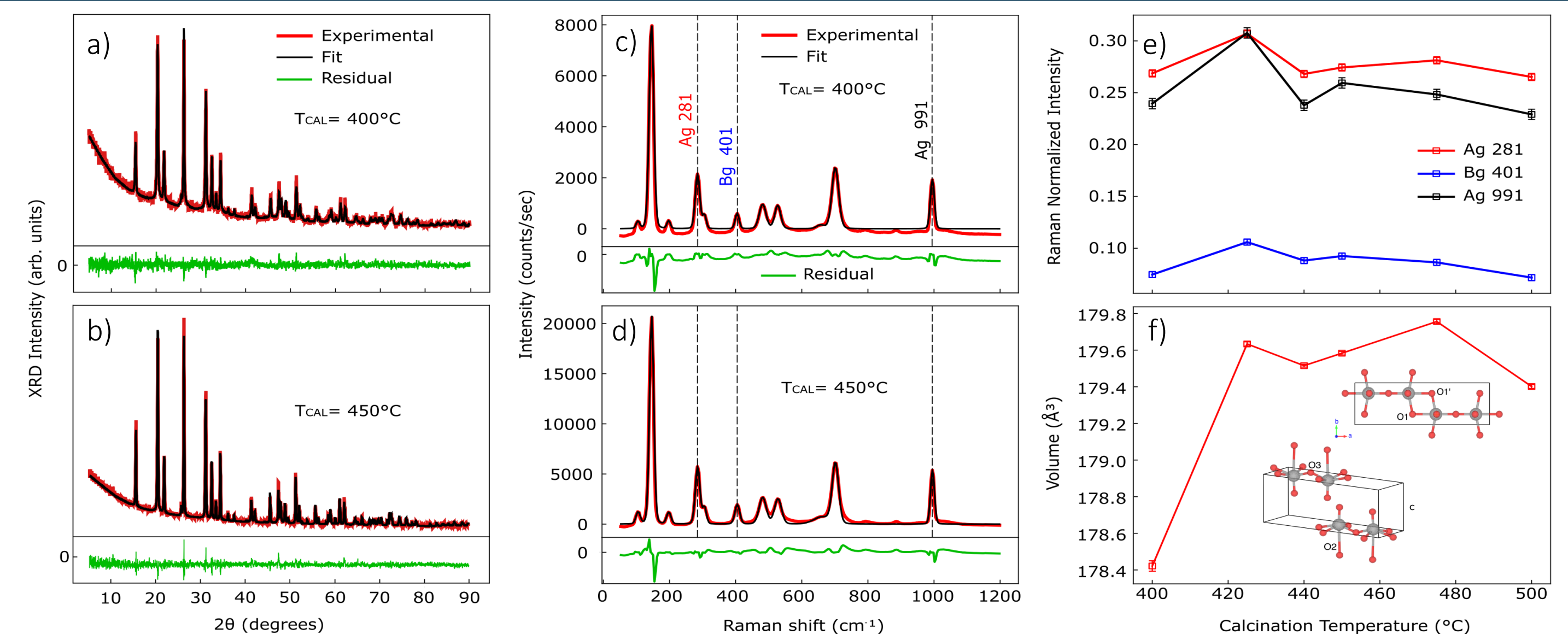


- Absorption edge shifts to lower photon energy decreasing non-monotonically with T_{cal} : E_g dips at 440°C and partially recovers at 450°C, comparable to the unit-cell-volume and the vanadyl raman intensities behaviour in [1] and our results.
- Tauc analysis of direct-transition gives total reduction of 0.52eV across 400°C-500°C (smaller than undoped [1]), suggesting vacancy-driven optical response is partially attenuated by Co-dopage.
- Direct vs. indirect exponents vary significantly, the transition character is not resolvable from optical data alone, should be settled against DFT-U band structure.



- Two linear regimes appear in the (n=2), a high-energy branch fit near the absorption onset (E_g^{high}) and a low-energy branch fit through the post-inflection region (E_g^{low})
- E_g^{high} is taken as the reported gap, since it fits to the region immediately adjacent to the onset, where the Tauc relation is derived to hold

Structural Analysis: XRD and Raman



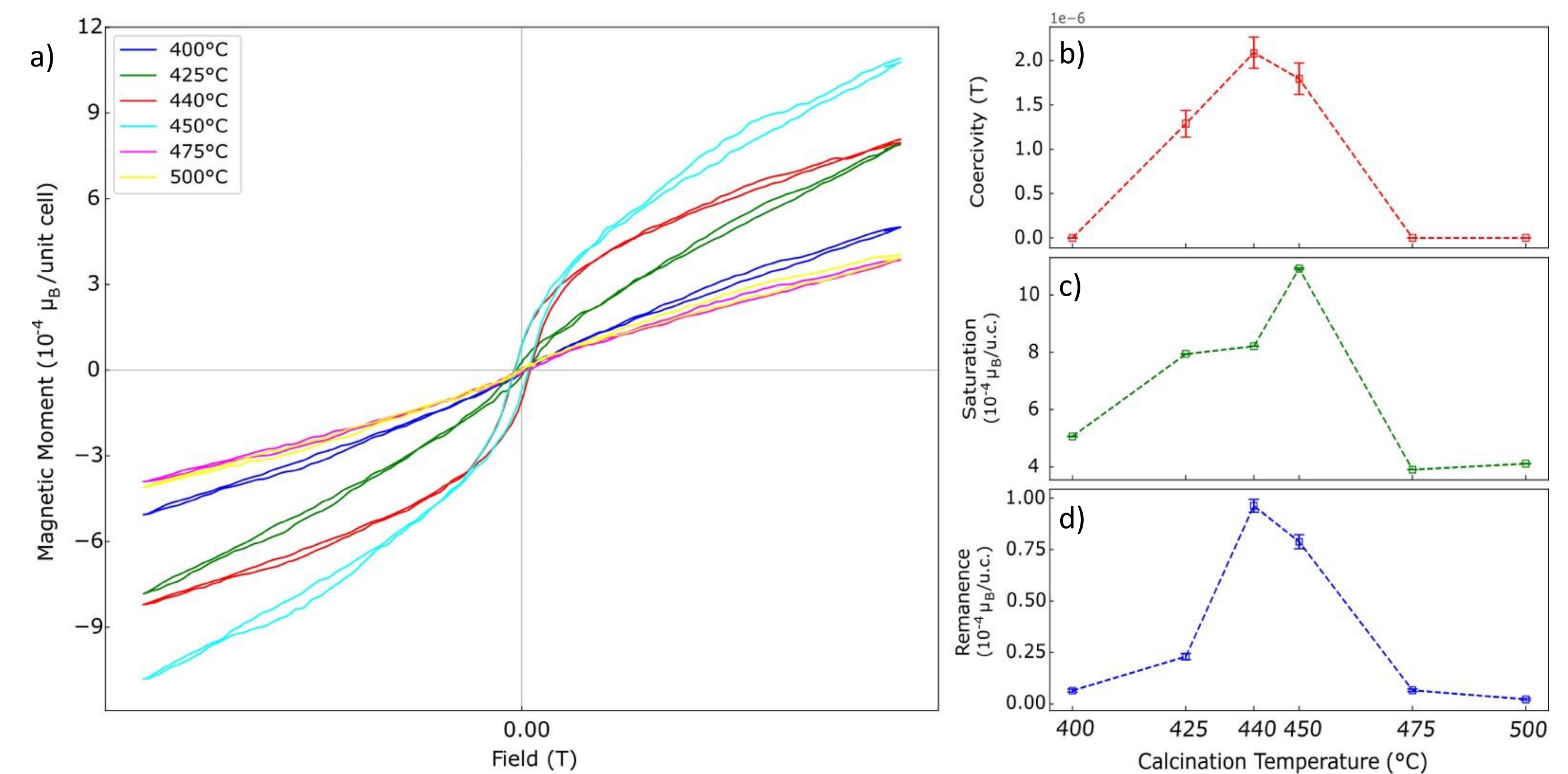
a-b) XRD intensity for $T_{cal} = 400^\circ\text{C}$ and 450°C .

c-d) Raman diffraction peaks labeling the three modes related to the V-O2, V-O3 and V=O1.

e) Normalized intensity of the peaks as a functions of T_{cal} . Ag-281 increases intensity compared to literature, strengthening the hypothesis of oxygen vacancies formation.

f) as expected, the volume tends to increase with calcination temperature, establishing a proportional relation between cell size volume and oxygen vacancy population.

Magnetic Properties



- T_{cal} controls the magnetic response by setting the concentration of O1 vacancies.
- Low T_{cal} (400–425 °C): Insufficient thermal energy for vacancy formation
No magnetic V^{4+} polarons → zero net magnetization.
- Optimal window (440–450 °C):
Maximum polaron formation → strongest magnetic signal.
- High T_{cal} (475–500 °C): Magnetic signature collapses completely. The mechanism will be addressed through DFT+U calculations.

Hypothesis: AFM Superexchange

- System Modeling:** DFT+U calculations on pristine, defective, and Co-doped V_2O_5 supercells isolate the atomic and electronic structure of the defect complex.
- Magnetic Ground State:** By calculating the energy difference between forced Ferromagnetic (FM) and Antiferromagnetic (AFM) configurations, we determine the true spin coupling between the Cobalt dopant and the V^{4+} polaron.
- Optical Gap Correlation:** High-resolution Density of States (DOS) mapping of the V_{3d} orbitals correlates the theoretical energy gap of localized V^{4+} states with experimental Tauc plots.

Conclusions

- Cobalt doping and thermal processing drive the formation of oxygen vacancies and localized V^{4+} polarons.
- While a 440–450 °C calcination window maximizes polaron formation, the net magnetization of the material is fundamentally suppressed
- The roadmap towards DFT calculations was laid out and will be implemented as the next step of this research. These calculations Will clear the origin of the magnetization suppression.

References:

[1] Correal, S., Hernández-Gómez, D., Esquivel, A. S., Cardona-Rodríguez, A., Reiber, A., Hernandez, Y., González-Hernández, R., & Ramírez, J. G. (2023). Tuning electronic and magnetic properties through disorder in V_2O_5 nanoparticles. Scientific Reports 2023 13:1, 13(1), 1–9. <https://doi.org/10.1038/s41598-023-32642-0>

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